Chapter - 2
Theory of Learning

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1 Plan

How to structure the chapter going forward.

Lecture 5 : Hypotheses can be infinitely many. But we need to first look at which are the relevant hypothesis that need to be counted = Dichotomies, Growth Function and Break Point.

Introduce only the concepts needed to understand how the number of effectively different classifications controls generalization. Use intuitive examples and figures; omit lengthy combinatorial derivations unless they are essential to the argument.

Lecture 6 : Through which we will arrive at the the VC inequality. Which is a modification to the Hoeffding Inequality that captures the real learning scenario.

Lecture 7 : Finally, we finish with the final theory notations of VC dimension and Generalization Bound.

Lecture 8 : might just add a few points from that.

2 References

1. CalTech Machine Learning Course - CS156, Lecture 5.
2. CalTech Machine Learning Course - CS156, Lecture 6.
3. CalTech Machine Learning Course - CS156, Lecture 7.
4. CalTech Machine Learning Course - CS156, Lecture 8.